	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University(SPPU)	

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Lab 3: Use case UML diagram

Title of the Case Study: TourMate – An Intelligent Tourism Management Platform

1. Aim:

To study and represent the functional requirements of the **TourMate** online tour management system using a **UML Use Case Diagram**, identifying all system actors, their roles, and the interactions between actors and the system through clearly defined use cases and relationships.

2. Actors and Their Roles

The actors in the TourMate Use Case Diagram are as follows:

1. **Tourist (End User):** This is the main actor for the system, and the tourists are the ones who will be interacting with the system for registration, login, viewing tour packages, booking, payment, viewing history, and feedback.

2. **Administrator:** This actor represents the person in charge of the entire system, handling everything within the system, from tour packages, customers, bookings, payment, hotels, and transport.

3. **Tour Operator/Agency:** This actor represents the secondary actor in the system, responsible for handling tour packages, scheduling a tour, and keeping track of the availability of existing tour packages. They operate in collaboration with the Administrator.

4. **Hotel/Transport Provider:** This actor represents the external service provider who offers availability information, handles bookings, and updates information regarding the service

provider's details, working in collaboration with the Administrator for hotel and transport management.

3. Use Case Packages

The use cases in the TourMate system are organized into the following functional packages:

1. User Access Handles the entry point of the system. Includes registration, login, and secure authentication. Both registration and login include secure authentication as a mandatory step.

- Register
- Login
- Secure Authentication *(included by Register and Login)*

2. Tour Browsing and Booking Covers the core functionality available to tourists for exploring and booking tours.

- Browse Tour Packages
- View Package Details (Destination, Price, Duration)
- Book Tour Package
- Make Payment *(included in booking)*
- Receive Booking Confirmation *(included in booking)*
- View Booking History

3. Feedback Allows tourists to engage with the system beyond booking.

- Submit Review and Rating *(extends Booking History)*
- View Personalized Suggestions *(extends Browse Tour Packages)*

4. Admin Management Covers all administrative operations within the system.

- Manage Tour Packages
- Manage Customer Records
- Approve / Modify Bookings *(extends Manage Tour Packages)*
- Manage Hotels and Transport
- Manage Payments
- Generate Reports
- Maintain System Security

5. Tour Operations Covers actions performed by Tour Operators.

- Create Tour Package
- Schedule Tours

- Update Tour Availability

6. Service Provider Covers actions performed by Hotels and Transport Providers.

- Provide Availability Information
- Confirm Bookings
- Update Services

4. Relationships in the Diagram

The Use Case Diagram uses three types of relationships:

Relationship	**Description**	**Example in TourMate**
Association	A direct link between an actor and a use case	Tourist → Book Tour Package
<<include>>	A use case that is always executed as part of another	Book Tour Package includes Make Payment
<<extend>>	A use case that optionally adds behavior to another	Submit Review extends View Booking History

5. Digram

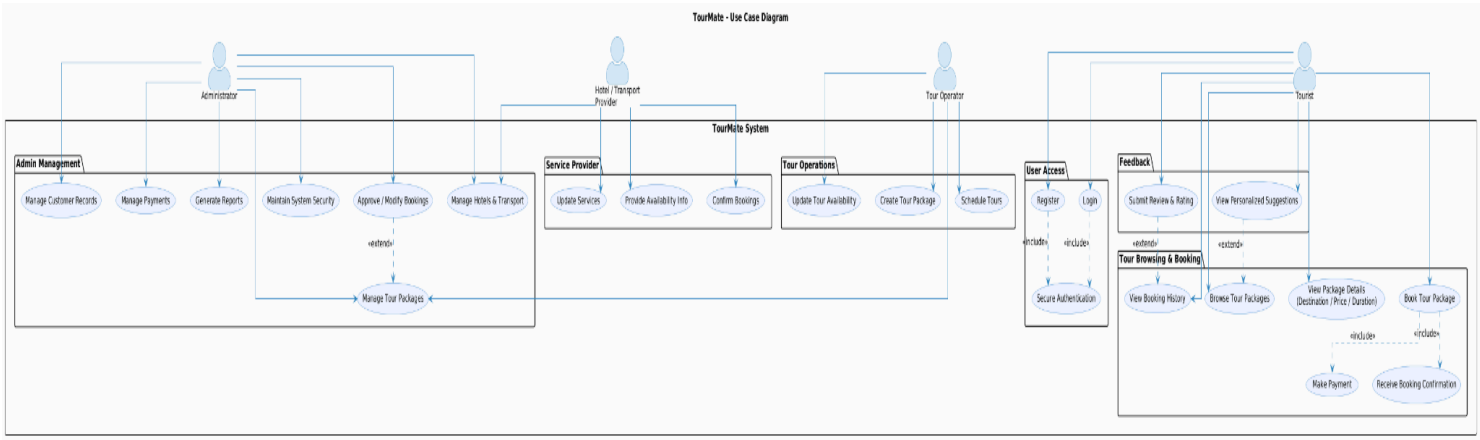

Fig.1 Usecase UML for Tourmate application

6. Conclusion

The Use Case Diagram for the TourMate system offers a complete overview of the interaction between the stakeholders and the system. It directly corresponds to the MoSCoW requirements as stated in Section 3, where the Must Have requirements are implemented, as well as the Could Have requirements. The Use Case Diagram is the basis upon which the Class, Sequence, and Activity diagrams are constructed.